SHANE WHELAN

Lucan, Dublin, Ireland
shane.whelan2@ucdconnect.ie · +353 83 048 7198
shane-whelan.ie · github.com/swhelan123 · linkedin.com/in/shane-whelan-364988291

Software Engineer — Electrical Captain @ UCD Formula Student

Education

BSc Computer Science with Data Science & AI 2023 – 2027 (Expected)
University College Dublin
GPA: 3.97/4.20, First Class Honours (1.1)

Leaving Certificate 2021 – 2023
Colaiste Phadraig CBS, Lucan
Achieved maximum possible points (625)

Skills

- **Languages:** Java, Python, C, C++, SQL, JavaScript, HTML/CSS, Bash
- **Backend & Infrastructure:** FastAPI, PostgreSQL (Supabase), WebSockets, Docker, Docker Compose, GitHub Actions, GitLab CI, Linux, Git
- **Embedded:** STM32, Arduino, CAN bus, SPI/isoSPI, I2C, UART, CMake, PlatformIO, custom bootloaders
- **Data:** pandas, NumPy, statsmodels, scikit-learn, SciPy, Matplotlib, Plotly, Streamlit, Hadoop MapReduce, MySQL
- **Testing & Tooling:** pytest, Unity (C), Artillery, Google Guava, PyQt6, LaTeX

Professional Experience

Software Engineer Intern June 2025 – August 2025
Twilio (Dublin / Remote)

- Led the deprecation of a legacy microservice whose APNs library was failing Twilio's 99.99% availability SLA, migrating its core logic into a high-throughput, real-time messaging platform.
- Built a new Apple Push Notification (APNs) binding adaptor in Java, piloting the `apns-http2` library to cut network errors and operational overhead.
- Designed a thread-safe, in-memory APNs client cache using Google Guava to improve performance and allow fair comparative load testing.
- Load tested the new implementation with Artillery, showing lower latency and higher throughput than the legacy service.

UCD Formula Student (UCDFS) Oct 2024 – Present
Electric race car built to compete at Formula Student UK, Silverstone. First Computer Science student on the team.

Electrical Captain (formerly Powertrain Captain) Jan 2026 – Present

- Lead a division of 19 engineers, renamed from Powertrain to Electrical for the 2026/27 season as its remit grew to cover the car's wider electronics. Led the division through FSUK 2026 at Silverstone.
- Remained the team's lead programmer. Wrote the firmware for our custom battery management system in C on an STM32F303: an LTC6812 isoSPI driver for a 75S7P accumulator, cell balancing, open-wire detection, fault handling, CAN, and a bootloader for in-field firmware updates.
- Built the BMS desktop tool in Python and PyQt6 for board bring-up, live cell voltage and temperature monitoring, configuration and firmware flashing, plus a protocol-level simulator with 12 fault modes so the whole stack could be tested before hardware arrived. Backed by ∼330 pytest and 181 C unit tests against mocked peripherals. *GitHub*

Powertrain Vice Captain Sep 2025 – Jan 2026

- Supported the captain on division decisions and project management, tracking progress and deadlines across the powertrain build while continuing as lead programmer.

Software Engineer, Vehicle Control Unit Oct 2024 – Sep 2025

- Wrote the Vehicle Control Unit firmware in C++ on a Teensy 4.1, controlling a Bamocar D3 motor controller and Emrax 208 motor over CAN, with torque mapping, regenerative braking and battery current limit handling. *GitHub*
- Implemented the FSUK safety logic: accelerator pedal plausibility checks, pedal/brake interlocks, IMD and BSPD fault monitoring, and a deceleration-triggered brake light.
- Integrated the BMS over CAN and drove a Nextion dashboard display for live telemetry.

Bar Staff 2025 – Present

Boland's Pub, Dublin (part-time, alongside full-time study)

Projects

UCDFS Team Platform – *ucdfs.shane-whelan.ie* · *GitHub* 2025 – Present

- Built and maintain the team's internal web platform in FastAPI and Supabase (PostgreSQL with row-level security): attendance, meetings, purchase requests, team directory and org chart, and role-based admin, behind httpOnly session auth.
- Wrote a real-time collaborative manufacturing planner over WebSockets, where members tick off build steps on a shared dependency graph; the powertrain plan tracked the car's build to 89% with a full audit log.
- Self-hosted with Docker on my home server. CI/CD runs on a self-hosted GitHub Actions runner: tests on every push, tagged images deployed automatically to staging, and approval-gated production releases with one-command rollback.

When Does Qualifying Position Matter in Formula 1? – *GitHub* Jan 2026 – Apr 2026

- Group data science project analysing six seasons (2020–2025) of race, qualifying and lap data collected with the FastF1 API.
- Designed a car-corrected Elo rating that adjusts for constructor qualifying gaps to separate driver skill from car performance.
- Led the joint OLS analysis across 131 races at 35 circuits, finding circuit layout was by far the strongest moderator of how much grid position decides the result, while driver skill and overtaking frequency were not significant.

S&P 500 Big Data Analysis – MapReduce vs. SQL – *GitHub* Jan 2026 – Mar 2026

- Processed 56.7 million rows (2.7 GB across 2,000 CSV files) of historical S&P 500 price data, benchmarking MySQL against distributed Hadoop MapReduce with custom Python mappers and reducers.

Home Server Ongoing

Run a Linux home server hosting ∼20 Docker containers, including the UCDFS platform, with self-hosted CI runners for GitHub Actions and GitLab.

Honours & Awards

Colm Church Scholarship Recipient 2026

One of 3 finalists selected by St. Raphael's Garda Credit Union for academic excellence and community contribution (Award: MacBook Air and annual educational bursary).

UCD Entrance Scholar Award 2023

Awarded to incoming students with the highest academic results from their secondary school.